



AI-driven performance in organizations: Unveiling the role of team dynamics, innovation, and commitment using a hybrid approach

Aziz Ullah^a, Xiaoming Sun^{a,*}, Wang Yalan^a, Adnan Ali^b

^a Management Science and Engineering, Xi'an University of Architecture and Technology, Xian, Shaanxi 710045, China

^b School of Management, Northwestern Polytechnical University, No. 3, Xigong Road, Xian, Shaanxi, China

ARTICLE INFO

Keywords:

Artificial intelligence
Team dynamics
Corporate innovation and innovation capabilities
High-commitment workforce

ABSTRACT

As artificial intelligence (AI) adoption accelerates across industries, the role of organizational mechanisms that enable its effective utilization becomes increasingly critical. Drawing on innovation diffusion theory, this study investigates how team dynamics, high-commitment workforces (HCW), and corporate innovation and innovation capabilities (CIIC) mediate the relationship between AI adoption and organizational performance. Using survey data from 419 Chinese manufacturing firms, the study employs structural equation modeling alongside long short-term memory techniques to validate the proposed framework. The findings reveal that cohesive team dynamics enhance collaboration and knowledge sharing, while strong CIIC promotes strategic alignment and innovation outcomes. Moreover, a highly committed workforce amplifies the performance benefits of AI adoption by fostering deeper employee engagement, greater adaptability, and enhanced organizational flexibility. This study contributes to the literature on organizational behavior and technology management by demonstrating how human and organizational enablers unlock the performance-enhancing potential of AI. The results also provide practical guidance for managers aiming to design supportive organizational environments and leverage AI to achieve sustainable competitive advantage.

1. Introduction

Artificial intelligence (AI) has emerged as a critical driver of organizational performance in the era of Industry 4.0, fundamentally reshaping how firms design operations, manage teams, and sustain competitiveness in technology-intensive environments (Gabsi, 2024; Khogali & Mekid, 2023). Historically, AI adoption in organizations was primarily associated with task automation and efficiency gains, focusing on cost reduction and process optimization (Ryalat, Franco, Elmoaqet, Almtireen, & Al-Refai, 2024). More recently, advances in machine learning, data analytics, and intelligent systems have expanded AI's role from operational support to strategic value creation, influencing innovation, decision-making quality, and organizational adaptability (Ali, Yang, Ali, Ali, & Noman, 2025a; Singh, Joshi, Dissanayake, Nainayake, & Kumar, 2025). For managers and policymakers, the practical challenge is no longer whether to adopt AI, but how to leverage it effectively to generate tangible performance gains while maintaining workforce engagement and organizational commitment. In manufacturing and related industries, AI applications such as the Internet of Things (IoT), predictive maintenance, real-time monitoring,

process automation, and demand forecasting enable firms to reduce operational costs, enhance flexibility, improve decision-making across supply chains, and support sustainability objectives by minimizing waste, optimizing resource use, and reducing energy consumption (Ali, Qian, Jiang, Fey, & Ali, 2026; Shamsuddoha et al., 2025). Beyond efficiency, AI facilitates mass customization and resilience in global value chains by allowing firms to respond rapidly to demand volatility and operational disruptions (Wan et al., 2020; Modgil, Singh, & Hannibal, 2022). As competitive pressures intensify, AI adoption has thus evolved from a purely technological upgrade into a strategic imperative with direct implications for organizational performance, innovation capacity, and long-term sustainability (Tariq, Poulin, & Abonamah, 2021).

Despite these practical promises, firms frequently struggle to translate AI investments into sustained performance improvements, highlighting a critical gap between adoption and value creation. Prior research offers limited insight into the organizational mechanisms through which AI-driven technologies enhance performance, often treating AI as a standalone technological input rather than a socio-technical system embedded within teams and organizational processes (Mikalef, Islam, Parida, Singh, & Altwaijry, 2023). Drawing on

* Corresponding author.

E-mail addresses: azizullah@xauat.edu.cn (A. Ullah), Sun413@xauat.edu.cn (X. Sun), wyl0622@xauat.edu.cn (W. Yalan), adnanali@nwpu.edu.cn (A. Ali).